

6	(a) (i) $2\pi f = 380$ frequency = 60 Hz	C1 A1	[2]
	(ii) $I_{\text{RMS}} \times \sqrt{2} = I_0$ $I_{\text{RMS}} = 9.9 / \sqrt{2}$ $= 7.0 \text{ A}$	C1 A1	[2]
	(b) power = $I^2 R$ $R = 400 / 7.0^2$ $= 8.2 \Omega$	C1 A1	[2]

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7 (a) wavelength of wave associated with a particle

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